

Section-by-Section Build Bible

How to build each part of the world + assets + scripts

Use this with your new **Universal 3D (URP)** project and the folder:

`Downloads\MyFantasyWorld_StarterPack\`

For every section below you get:

1. **What you are building**
 2. **Click steps (go here → do this)**
 3. **How to color / paint the ground**
 4. **Assets you need** (included in pack vs download)
 5. **Scripts to add** (if any) + how to attach them
 6. **Done checklist**
-

How to import scripts once (do this before Section 1)

1. Create folder in Unity: `Assets/MyWorld/Scripts`
2. Copy all folders from pack `Scripts/` into it (Core, Player, Interaction, Buildings, Vehicles, World)
3. Wait until Unity finishes compiling
4. Create Tag: **Player**
(Edit → Project Settings → Tags and Layers)

Full script details also in: `Docs/HOW_TO_USE_SCRIPTS.md`

SECTION 0 — New project + empty scene

What you are building

A clean URP project and saved world scene.

Click steps

1. Unity Hub → **Projects** → **New project**
2. Select **Universal 3D** (URP)
3. Name it `MyFantasyWorld` → Create
4. File → Save As → `Assets/Scenes/MainWorld.unity`
5. Hierarchy empty → Create Empty → name `World`
6. Under World create empties: `TerrainRoot`, `Zones`, `Water`, `Roads`, `Props`, `Lighting`, `Systems`

Ground / assets / scripts

- Assets: none yet
- Scripts: none yet

Done

- ☐ URP project open
- ☐ MainWorld scene saved
- ☐ World folders exist

SECTION 1 — Terrain (the ground)

What you are building

Walkable land you can sculpt and paint.

Click steps

1. Right-click `TerrainRoot` → **3D Object** → **Terrain**
2. Select Terrain → Inspector → Terrain Settings (gear)
3. Set Width/Length `500`, Height `120`, Heightmap Resolution `513`
4. Press **F** in Scene view to frame it

How to color ground

Not yet — first shape land in Section 2, then paint in Section 3.

Assets needed

Included in pack	Download if missing
<code>AssetsToImport/Game_Terrains</code>	Extra terrain tools (optional)
<code>AssetsToImport/Game_Textures</code>	More PBR ground packs

Scripts

None on Terrain itself.

Done

- ☐ Terrain visible
- ☐ Size set
- ☐ Scene saved (Ctrl+S)

SECTION 2 — Hills & mountains (shape the land)

What you are building

Beach flat, village plateau, forest rise, river trench, mountains.

Click steps

1. Select Terrain → **Paint Terrain** → **Raise or Lower Terrain**
2. Brush Size 40–80, Opacity 0.05–0.2 (keep low)
3. Raise soft hills inland
4. Raise taller mountains at the north
5. Switch to **Smooth Height** and smooth slopes
6. Use **Flatten / Set Height** for beach (low) and village (a bit higher)
7. Use **Lower Terrain** for a winding river trench

How to color ground

Still later (Section 3). Shape first.

Assets / scripts

None required.

Done

- ☐ Beach flat exists
- ☐ Village flat exists
- ☐ Mountains readable
- ☐ River trench carved

SECTION 3 — Paint / color the ground (textures)

What you are building

Sand, grass, dirt, forest floor, rock colors on terrain.

Click steps

1. Copy AssetsToImport/Game_Textures into project Assets/
2. Select Terrain → **Paint Terrain** → **Paint Texture**
3. **Edit Terrain Layers** → **Create Layer** (or Add Layer)
4. Assign Diffuse (and Normal if you have it)
5. Set Layer Size about 10–20
6. Make layers: Sand, Grass, Dirt, ForestFloor, Rock
7. Paint base grass on meadows
8. Paint sand on beach
9. Paint dirt where paths will go
10. Paint rock on steep mountain sides
11. Soft-blend borders with low opacity

Assets needed

Included	Download
Game_Textures, Game_Terrains layers	Snow/mud/swamp textures later
Free Island textures (if imported)	Search: terrain textures PBR

Scripts

None.

Done

- ☐ At least 4 terrain layers
- ☐ Beach reads as sand
- ☐ Mountains have rock

SECTION 4 — Water (ocean / lake / river)

What you are building

Water surface at correct height.

Click steps

1. Right-click `Water` → **3D Object** → **Plane** → name `OceanWater`
2. Scale large enough for coast
3. Move Y to beach sea level
4. Create Material `M_Water` (URP Lit), blue, high Smoothness → drag onto plane
OR use prefab from `AQUAS-Lite` / `Phantom_Water`
5. Duplicate for `RiverWater` / `LakeWater` and fit into trenches/bowls

How to color ground near water

Paint sand/mud at shoreline; blend to grass inland.

Assets needed

Included	Download
<code>AQUAS-Lite</code> , <code>Phantom_Water</code>	Better <code>stylized water URP</code> (recommended)

Scripts

Script	When
<code>BoatController</code>	Only if you add boats (Section 16)
Set <code>Water Level Y</code> on boat to this water height	

Done

- ☐ Water touches shoreline
- ☐ Not floating high above land

SECTION 5 — Beach zone

Zone empty

Create `Zones/Zone_Beach`

Click steps

1. Flatten south strip + sand paint (already partly done)
2. Import pier from `Game_Models/wooden-pier` (if present)
3. Place pier under `Zone_Beach`
4. Place 1–3 boats from `Alstra_Boats`
5. Add a few rocks from `Free_Rocks`
6. Optional palms (download)

Ground color

Sand near water → blend to grass inland.

Assets needed

Included	Download
Pier, boats, rocks, sand textures	Palm trees URP, better beach props

Scripts

Script	How
<code>ZoneTrigger</code>	Empty + Box Collider (Is Trigger) sized over beach → add script → names <code>Zone_Beach</code> / <code>Beach</code>
<code>AmbientZoneAudio</code>	Same zone or child → Audio Source with waves loop → add script
<code>BoatController</code> + enter/exit	On boat (Section 16)

Done

- [] Beach looks like shore
- [] Path leaving beach inland

SECTION 6 — Forest zone (paint many trees)

Zone empty

`Zones/Zone_Forest`

What you are building

Dense forest by **painting trees**, not placing one-by-one.

Click steps

1. Import `ALP_Assets`, `Forst`, `Game_RocksAndVegetation`
2. For `Forst`: open pack → double-click `Conifers_URP.unitypackage` → Import

3. Select Terrain → toolbar **Paint Trees**
4. **Edit Trees** → **Add Tree**
5. Drag a tree prefab into Tree Prefab → Add
6. Add 2–3 tree types (oak/maple + pine)
7. Brush over forest area with medium density
8. Leave openings for paths and clearings
9. Hand-place 2–3 “hero” trees under `Zone_Forest` for special spots

How to color ground in forest

Paint darker grass / forest floor texture. Less sand. Soft dirt path through trees.

Assets needed

Included	Download
ALP oak/poplar, Forst conifers, maple/fern pack	Jungle/palm packs for tropical forest
BushGrassBend (tool)	Extra bush packs

Scripts

Script	How
<code>ZoneTrigger</code>	Forest volume trigger
<code>AmbientZoneAudio</code>	Wind/birds clip
No tree script needed	Unity Terrain handles painted trees

Done

- ☐ Forest readable from distance
- ☐ Paths not blocked by trees
- ☐ Clearings exist

SECTION 7 — Grass & flowers (meadow)

Zone empty

`Zones/Zone_Meadow`

Click steps

1. Terrain → **Paint Details**
2. Edit Details → **Add Grass Texture** (from ALP GrassFlowers) or Detail Mesh (fern/bush)
3. Paint meadow heavily
4. Paint forest lightly
5. Almost no grass on pure sand or steep rock

Ground color

Bright grass TerrainLayer under meadow.

Assets needed

Included	Download
ALP GrassFlowers, ferns/bushes	More flower detail packs

Scripts

None required (optional meadow ZoneTrigger + birds audio).

Done

- [] Meadow feels soft and open

SECTION 8 — Rocks & cliffs

Click steps

1. Import `Free_Rocks` + rock prefabs from vegetation pack
2. Place under `Zone_Mountain` / river banks
3. Rotate (E) and scale (R) randomly
4. Sink slightly into terrain
5. Paint rock TerrainLayer under them

Assets needed

Included	Download
Free_Rocks, Game stone/desert rocks	Bigger cliff megapack

Scripts

None.

Done

- [] No floating rocks

SECTION 9 — Paths & roads

Click steps (easiest first)

1. Paint **Dirt** TerrainLayer as a winding path: Beach → Village → Forest → River → Mountain

Better roads later

1. Import `EasyRoads3D`

2. Create EasyRoads road network (GameObject menu → EasyRoads3D)
3. Shift+click path points
4. Build so road sits on terrain
5. Parent under `Roads`

Ground color

Dirt under/around road always — even if mesh road exists.

Assets needed

Included	Download
EasyRoads3D, KajamansRoads	Medieval/dirt road materials

Scripts

None required.

Done

- [] You can walk a clear route through the valley

SECTION 10 — River + bridge

Zone empty

`Zone_River`

Click steps

1. River trench + river water already placed
2. Import `3DWorldInSeconds_Bridges`
3. Drag wooden/stone bridge prefab across river
4. Align deck with path height
5. Add rocks under supports
6. Continue dirt path both sides

Ground color

Rock/moss near water; dirt on approach.

Assets needed

Included	Download
Bridges pack	Extra fantasy bridges

Scripts

Script	How
ZoneTrigger	Optional river zone
AmbientZoneAudio	Water flow sound

Done

- [] Player can cross bridge

SECTION 11 — Fantasy village (houses)

Zone empty

Zone_Village

Click steps

1. Import 3DForge_FantasyExteriors
2. Flatten village plateau; paint dirt + grass + plaza
3. Place path into village first
4. Drag houses/fences/farm pieces under Zone_Village
5. Face doors toward roads
6. Add well/sheds from Game_Models if present
7. When a house looks good: drag to Assets/Prefabs/Buildings/

Ground color

Dirt roads, grass yards, optional paving for square.

Assets needed

Included	Download
3DForge village kit, sheds/well	Market stalls, interior furniture, more props

Scripts — HOUSE DOOR

1. On house, create empty Door_Interact at door
2. Add collider
3. Add script HouseDoor
4. Create InteriorSpawn inside and ExteriorSpawn outside
5. Assign both on HouseDoor
6. Mode: **Teleport To Interior** (or Toggle Door Rotation for swing doors)
7. Optional: BuildingInteriorVolume trigger box inside house

Player must have PlayerInteraction (Section 12) to press E.

Done

- [] 3–6 houses placed

- [] At least one door interactable

SECTION 12 — Player (walk the world)

Click steps

1. Create Capsule (or character model) → Tag **Player**
2. Add **Character Controller**
3. Add **PlayerMotor**
4. Add **PlayerInteraction**
5. On Main Camera add **SimpleFollowCamera** → assign Player
6. Create **PlayerSpawn** with **PlayerSpawnPoint** on beach/village
7. Empty **Systems** + **PlayerBootstrap** → assign PlayerMotor
8. Set PlayerMotor Ground Mask to include Default/Terrain
9. Press Play — WASD walk, Shift run, Space jump, E interact

Assets needed

Included	Download
Scripts in pack; optional StarterAssets	Fantasy hero mesh / Starter Assets Third Person

Scripts used

PlayerMotor, **PlayerInteraction**, **PlayerSpawnPoint**, **PlayerBootstrap**, **SimpleFollowCamera**

Done

- [] Can walk beach → village without falling

SECTION 13 — Farm & countryside

Zone empty

Zone_Farm

Click steps

1. Flatten farm fields near village
2. Paint dirt strips + grass borders
3. Place 3DForge fences, stables, troughs
4. Keep path from village to farm clear

Ground color

Dirt rows + grass edges.

Assets needed

Included	Download
3DForge farm pieces	Crop meshes, farm animals

Scripts

Optional `ZoneTrigger`. Animals later (download + place; no required script yet).

Done

- [] Farm readable as farmland

SECTION 14 — Town square / market

Click steps

1. Flatten plaza in village center
2. Paint paving/dirt plaza
3. Place well / center prop
4. Place stalls/barrels/crates (download props if missing)

Assets needed

Included	Download
Partial village props	medieval props , market stall

Scripts

None required. NPCs later (Section 20).

SECTION 15 — Cars

Click steps

1. Import a car from `Game_Models` (Jeep etc.) or download car model
2. Root: **Rigidbody** mass 1200–1600
3. Add 4 children **WheelCollider** at wheels (`WC_FL/FR/RL/RR`)
4. Add `CarController` → assign wheel colliders (+ visual wheels optional)
5. Add empty COM below center → assign Center Of Mass
6. Add `VehicleSeat` + `VehicleEnterExit`
7. Create `ExitPoint` beside car → assign on seat
8. Play: walk up, press **E**, drive WASD, **E** exit

Ground color

Keep roads dirt/gravel under car routes.

Assets needed

Included	Download
Game car models (meshes)	Cleaner wheel-ready cars if needed

Scripts

CarController, VehicleSeat, VehicleEnterExit

Requires player scripts from Section 12.

Done

- [] Enter/drive/exit works on flat ground

SECTION 16 — Bikes

Click steps

1. Place bike model from Game_Models or download
2. Rigidbody mass 200–280
3. 2 WheelColliders (front/rear)
4. Add BikeController + Seat + EnterExit
5. Test on flat ground first

Assets / scripts

Bike model + BikeController + enter/exit scripts.

SECTION 17 — Planes

Click steps

1. **Download an airplane model** (usually not in pack)
2. Add Rigidbody + PlaneController + Seat + EnterExit
3. Place on flat beach/runway strip
4. Controls: W thrust, A/D yaw, Q/E roll, R/F pitch, E exit

Assets needed

Included	Download
Scripts only	Airplane model URP (required)

SECTION 18 — Boats

Click steps

1. Place boat from Alstra_Boats
2. Add Rigidbody + BoatController

3. Set **Water Level Y** = your ocean/lake Y
4. Add Seat + EnterExit
5. Test near pier

Assets / scripts

Boats included + `BoatController`.

SECTION 19 — Lakes, waterfall, cave mouth

Lakes

Lower a bowl → lake water plane → mud/rock shoreline paint.

Waterfall

Cliff shelf + waterfall VFX (download) + water audio.

Cave

Rock mouth arrangement under `Zone_Cave` (full cave mesh later/download).

Scripts

ZoneTrigger + AmbientZoneAudio (water/wind).

Download

Waterfall VFX, cave pack.

SECTION 20 — Life: audio, day/night, zones, NPCs

Day / night

1. Empty `DayNight` under Systems
2. Add `DayNightCycle`
3. Assign Directional Light

Zone audio

1. On each zone trigger object add Audio Source (loop clip)
2. Add `AmbientZoneAudio`
3. Use waves/wind/birds clips (**download audio packs**)

Zone names

Add `ZoneTrigger` boxes for Beach/Village/Forest/Mountain.

NPCs / animals

Download villagers/birds/animals → place under zones (no AI required at first).

Torches / campfires

Download torch/campfire VFX → place + Point Light for night.

Assets to download

Nature ambience audio, rain VFX, villagers, birds, torch/campfire.

Scripts

DayNightCycle, AmbientZoneAudio, ZoneTrigger

SECTION 21 — Extra biomes (later)

Biome	Ground color	Assets	Scripts
Snow peaks	Snow layer on tops	Snow textures + pines	ZoneTrigger
Desert	Sand + sparse rock	Desert rocks (included partial)	ZoneTrigger
Swamp	Mud + moss + shallow water	Swamp trees (download)	Fog volume + audio
Jungle	Dark ground + dense paint trees	Jungle/palm download	Zone audio

SECTION 22 — Sky, sun, fog (look)

Click steps

1. Import AllSkyFree
2. Window → Rendering → Lighting → Environment → Skybox Material
3. Rotate Directional Light for nice sun angle
4. Lighting empty → Volume → Global Volume → add Fog / Bloom (subtle)

Assets

AllSkyFree included. Optional more skies.

Scripts

Works with DayNightCycle.

MASTER ORDER (do in this sequence)

1. Section 0 Project
2. Section 1–3 Terrain + paint
3. Section 4–5 Water + beach
4. Section 12 Player (so you can walk while building)
5. Section 9 Path

6. Section 11 Village + HouseDoor script
7. Section 6–8 Forest, grass, rocks
8. Section 10 Bridge
9. Section 13–14 Farm + market
10. Section 15–18 Vehicles you care about
11. Section 19–22 Life + sky + biomes

Quick script cheat sheet by section

Section	Scripts to add
Terrain / paint / trees / grass / rocks / roads	None
Beach / forest / river zones	ZoneTrigger , AmbientZoneAudio
House	HouseDoor , optional BuildingInteriorVolume
Player	PlayerMotor , PlayerInteraction , spawn/bootstrap, camera
Car	CarController + VehicleSeat + VehicleEnterExit
Bike	BikeController + seat + enter/exit
Plane	PlaneController + seat + enter/exit
Boat	BoatController + seat + enter/exit
Day/night	DayNightCycle
Whole world feel	Zone + audio scripts on each biome

If something is missing

Open also:

- Docs/HOW_TO_USE_SCRIPTS.md — deeper script setup
- Docs/ASSETS_INCLUDED_AND_DOWNLOAD.md — full download list
- Docs/FROM_SCRATCH_WORLD_BUILDING_GUIDE.md — longer beginner theory
- HowToImport/IMPORT_ORDER.txt — import order for art packs

Rule: Each section = build steps + ground color + assets + scripts.
Finish one section's checklist before jumping ahead.